

Jingyu (Jack) Zhang

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RESEARCH INTERESTS

My research interest lies in the area of natural language processing. I am particularly interested in the **responsible development and deployment of foundation models**. My recent work focuses on safety alignment and enhancing verifiability of LLMs.

EDUCATION

Johns Hopkins University

Aug 2023 - Present

Ph.D. in Computer Science

Advisors: Daniel Khashabi and Benjamin Van Durme

Johns Hopkins University

Aug 2019 - May 2023

B.S. in Computer Science

GPA: 3.97/4.00

Additional Majors: Mathematics; Applied Mathematics & Statistics. Minor: Economics

PUBLICATIONS & PREPRINTS

Selected Works.....

1. **Controllable Safety Alignment: Inference-Time Adaptation to Diverse Safety Requirements**

Jingyu Zhang, Ahmed Elgohary, Ahmed Magooda, Daniel Khashabi, Benjamin Van Durme.

ICLR 2025

2. **Verifiable by Design: Aligning Language Models to Quote from Pre-Training Data**

Jingyu Zhang, Marc Marone, Tianjian Li, Benjamin Van Durme, Daniel Khashabi.

NAACL 2025

3. **SemStamp: A Semantic Watermark with Paraphrastic Robustness for Text Generation**

Abe Bohan Hou*, **Jingyu Zhang***, Tianxing He*, Yichen Wang, Yung-Sung Chuang, Hongwei Wang, Lingfeng Shen, Benjamin Van Durme, Daniel Khashabi, Yulia Tsvetkov.

NAACL 2024. [*Equal Contribution]

4. **On the Blind Spots of Model-Based Evaluation Metrics for Text Generation**

Tianxing He*, **Jingyu Zhang***, Tianle Wang, Sachin Kumar, Kyunghyun Cho, James Glass, Yulia Tsvetkov.

ACL 2023. **Oral Presentation**. [*Equal Contribution]

All Publications.....

1. **Rationalyst: Pre-training Process-Supervision for Improving Reasoning**

Dongwei Jiang, Guoxuan Wang, Yining Lu, Andrew Wang, **Jingyu Zhang**, Chuyu Liu, Benjamin Van Durme, Daniel Khashabi.

arXiv preprint: <https://arxiv.org/abs/2410.01044>

2. **Core: Robust Factual Precision Scoring with Informative Sub-Claim Identification**

Zhengping Jiang, **Jingyu Zhang**, Nathaniel Weir, Seth Ebner, Miriam Wanner, Kate Sanders, Daniel Khashabi, Anqi Liu and Benjamin Van Durme.

arXiv preprint: <https://arxiv.org/abs/2407.03572>

3. **Self-(In)Correct: LLMs Struggle with Refining Self-Generated Responses.**

Dongwei Jiang, **Jingyu Zhang**, Orion Weller, Nathaniel Weir, Benjamin Van Durme, Daniel Khashabi.

AAAI 2025

4. **TurkingBench: A Challenge Benchmark for Web Agents**

Kevin Xu, Yeganeh Kordi, Kate Sanders, Yizhong Wang, Adam Byerly, **Jingyu Zhang**, Benjamin Van Durme, Daniel Khashabi.

NAACL 2025

5. **DiffNorm: Self-Supervised Normalization for Non-autoregressive Speech-to-speech Translation.**

Weiting Tan, **Jingyu Zhang**, Lingfeng Shen, Daniel Khashabi, Philipp Koehn.

NeurIPS 2024.

6. **k-SemStamp: A Clustering-Based Semantic Watermark for Detection of Machine-Generated Text**

Abe Bohan Hou, **Jingyu Zhang**, Yichen Wang, Daniel Khashabi, Tianxing He.

Findings of ACL 2024.

7. **The Language Barrier: Dissecting Safety Challenges of LLMs in Multilingual Contexts**
Lingfeng Shen, Weiting Tan, Sihao Chen, Yunmo Chen, Jingyu Zhang, Haoran Xu, Boyuan Zheng, Philipp Koehn, Daniel Khashabi.
Findings of ACL 2024.
8. **On the Zero-Shot Generalization of Machine-Generated Text Detectors**
Xiao Pu, Jingyu Zhang, Xiaochuang Han, Yulia Tsvetkov, Tianxing He.
Findings of EMNLP 2023.
9. **Geo-Seq2seq: Twitter User Geolocation on Noisy Data through Sequence to Sequence Learning**
Jingyu Zhang, *Alexandra DeLucia, Chenyu Zhang, Mark Dredze.*
Findings of ACL 2023.
10. **PCFG-based Natural Language Interface Improves Generalization for Controlled Text Generation**
Jingyu Zhang, *James Glass, Tianxing He.*
**SEM 2023. Preliminary version accepted at 2nd Workshop on Efficient Natural Language and Speech Processing (ENLSP), NeurIPS 2022. Best Paper Award.*
11. **Changes in Tweet Geolocation over Time: A Study with Carmen 2.0**
Jingyu Zhang, *Alexandra DeLucia, Mark Dredze.*
8th Workshop on Noisy User-generated Text (W-NUT), COLING 2022.
12. **Study of Manifestation of Civil Unrest on Twitter**
Abhinav Chinta, Jingyu Zhang*, Alexandra DeLucia, Anna L. Buzcak, Mark Dredze.*
*7th Workshop on Noisy User-generated Text (W-NUT), EMNLP 2021. [*Equal Contribution]*

RESEARCH EXPERIENCE

Microsoft Responsible AI Research <i>Research Intern</i> Mentor: Ahmed Elgohary Ghoneim	<i>May 2024 - Aug 2024, P/T Sep 2024 - Present</i>
Center for Language and Speech Processing at Johns Hopkins University <i>PhD Researcher</i> Advisors: Daniel Khashabi and Benjamin Van Durme	<i>Fall 2023 - Present</i>
University of Washington <i>Undergraduate Research Intern</i> Advisors: Yulia Tsvetkov and Tianxing He	<i>Summer 2022 - Spring 2023</i>
MIT Computer Science and Artificial Intelligence Laboratory <i>Undergraduate Research Intern</i> Advisors: James Glass and Tianxing He	<i>Spring 2022 - Summer 2022</i>
Center for Language and Speech Processing at Johns Hopkins University <i>Undergraduate Researcher</i> Advisors: Mark Dredze and Benjamin Van Durme	<i>Spring 2021 - May 2023</i>

AWARDS & HONORS

- **Best Paper Award** - ENLSP Workshop at NeurIPS 2022
- **Michael J. Muuss Research Award** - 1 out of 744 JHU CS undergraduates (2023). Press coverage: [🔗](#)
- **CRA Outstanding Undergraduate Researcher Award Nominee** - 4 out of 744 JHU CS undergraduates (2022)
- **Pistrutto Research Fellowship** - \$4000 grant (Fall 2022). Press coverage: [🔗](#)
- **Bloomberg Distinguished Professor (BDP) Summer Program Recipient** - \$6000 grant (Summer 2021).
- **Upsilon Pi Epsilon** - International Honor Society for the Computing and Information Disciplines.
- **National Olympiad in Informatics in Provinces (NOIP)** - National 1st Prize Certification (2018)

TALKS

Controllable Safety Alignment: Inference-Time Adaptation to Diverse Safety Requirements <i>Invited talk, CLSP Seminar, Johns Hopkins University, Baltimore, Maryland</i> <i>Invited talk, Microsoft Responsible AI Tech Talks, Redmond, Washington</i>	<i>Oct 2024</i> <i>Dec 2024</i>
Verifiable by Design: Aligning Language Models to Quote from Pre-Training Data <i>Oral presentation, the 11th Mid-Atlantic Student Colloquium on Speech, Language and Learning</i> <i>Invited talk, Microsoft Hallucination Deep Dive, Redmond, Washington</i> <i>Invited talk, Institute for Assured Autonomy, Johns Hopkins University, Baltimore, Maryland</i>	<i>May 2024</i> <i>Aug 2024</i> <i>Oct 2024</i>
SemStamp: A Semantic Watermark with Paraphrastic Robustness for Text Generation	

Invited talk, CLSP Seminar, Johns Hopkins University, Baltimore, Maryland

Dec 2023

On the Blind Spots of Model-Based Evaluation Metrics for Text Generation

Oral Presentation, ACL 2023, Toronto, Canada

July 2023

PCFG-based Natural Language Interface Improves Generalization for Controlled Text Generation

Spotlight Presentation, ENLSP Workshop, NeurIPS 2022, New Orleans, USA

Dec 2022

INDUSTRY EXPERIENCE

Microsoft

Responsible AI Research

Research Intern

May 2024 - Present (P/T after Aug 2024)

- Conducting research related to safety alignment and automatic red teaming
- Resulting in a ICLR 2025 paper, available at <https://arxiv.org/abs/2410.08968>.

ByteDance Ltd.

Lark Explorer Department

C++ Development Intern

May 2020 - May 2021 (P/T after Aug 2020)

- Carried out C++ cross-platform development interacting with macOS kernel and Windows Win32 API
- Performed client-side development with Electron and Node.js related to performance optimization
- Conducted data science analytics on extensive user-generated data with Apache Hive and Python

TEACHING EXPERIENCE

EN.601.465/665 Natural Language Processing

JHU Department of Computer Science

Course Assistant

Fall 2021, Fall 2022

- EN.601.465/665 is a mixed graduate / upper-level undergraduate course in NLP taught by Prof. Jason Eisner
- Conducted grading of homework and exam papers and held review sessions and office hours on a weekly basis
- Received an average score of 4.80/5.00 on student TA evaluation (100% “Good” or “Excellent” rating)

Code in Place 2021

Stanford University Department of Computer Science

Section Leader (Volunteer)

April 2021 - May 2021

- Worked with a team of more than 50 teaching leads and 1000 section leaders to support 10,000+ students across the world as they navigate the first five weeks of CS 106A: Programming Methodology course
- Prepared materials and taught a Python programming section of 10 students on a weekly basis

ACADEMIC SERVICE

- Application Mentor, JHU CLSP pre-application support program (2023-present)
- Curriculum Committee, Department of Computer Science, Johns Hopkins University (2023-present)
- Recruitment Committee, Center for Language and Speech Processing, Johns Hopkins University (2023-present)

SKILLS & COURSEWORK

Technical Skills

- Programming Languages: Python, C/C++, Java, OCaml, HTML, CSS, Javascript, MATLAB, SQL
- Frameworks: Huggingface, PyTorch, Sklearn, Pandas, Numpy, Electron, Windows/macOS native APIs
- Development Workflow: L^AT_EX, Bash, Emacs, Git, Makefile, GN build

Related Coursework

- NLP/ML: Natural Language Processing, Machine Translation, Multilingual NLP, Artificial Agents, Machine Learning, Deep Learning, Human-Computer Interaction, Probabilistic Models of the Visual Cortex
- Math/Stats: Real Analysis, Abstract Algebra, Topology, Differential Equations, Probability, Statistics, Optimization, Time Series Analysis, Game Theory, Mathematical & Computational Foundations of Data Science